

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Designed the order book with `std::list` FIFOs per price level in a `std::map` and an order-ID hash map to make cancels $O(1)$, which ran the full replay at **1.26M events/sec** and hit a 1.2 μ s p99 on a separate 2M-event run
- Implemented B+ tree and LSM-tree backends for the engine's event log to compare them on identical 2M-record workloads, which showed the B+ tree giving **12.8x** the read throughput and 22x faster range queries for 27% more disk
- Traced an apparent 1.65x regression to the CPU frequency governor with perf counters and a do-nothing control book to rule out a code change, which led to core pinning that cut run-to-run variance from 6% to 0.06%

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Cut AI workflow generation time **65%** to under 15s by splitting a Claude skill into focused sub-skills that call Python search and validation tools instead of relying on open-ended model search
- Built the Claude skill's main search tool by precomputing popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Built and prompt-tuned a pipeline for a PhD-led robotics project aimed at ICLR that overlays AnyGrasp's candidate grasps on camera images in Open3D to help a vision-language model pick grasps that avoid hitting the table

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Rewrote Tortuga's Prisma ORM queries and added Redis caching to take load off compute, which kept the class scheduler from crashing through 20,000+ visits in its first 30 days

Technical Skills

Languages: C++17, C, Python, Java, Rust, SQL, TypeScript, Bash

Systems & Performance: debugging and profiling with Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, data structures and algorithms, benchmarking

Cloud & Tools: AWS (Lambda, Kinesis, EC2, Fargate, CloudWatch monitoring), AWS CDK, PostgreSQL, Redis, Docker, GitHub Actions (CI/CD), Git, unit testing (Pytest)

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)